EX 4

Structural design III - HS 2022

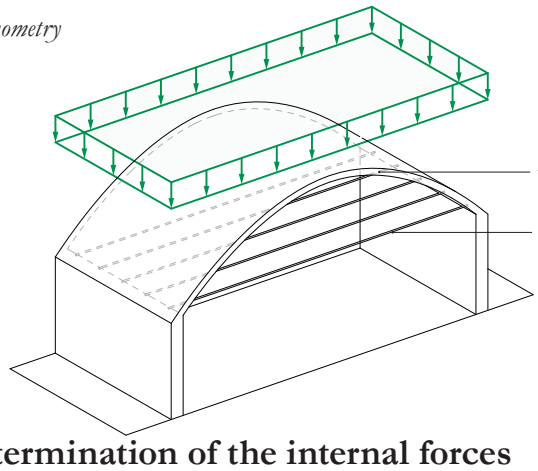
Analysis and development of structures in masonry

Name: _____

Task 1 Tranfer of vertical loads: design variation 1

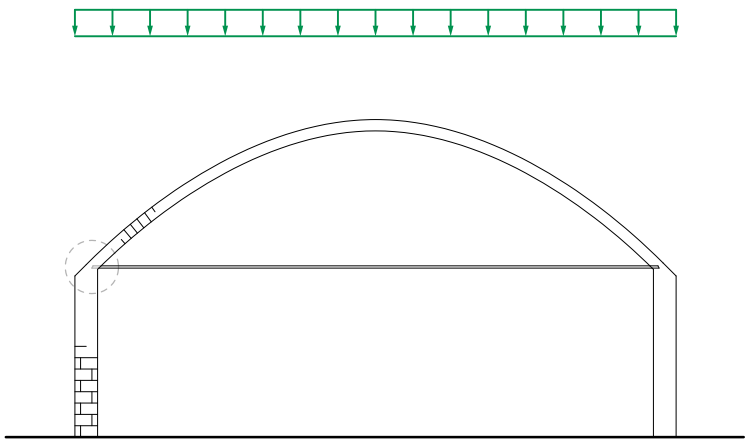
Structure equilibrating the horizontal thrust by means of tension cables

Axonometry



1a Determination of the internal forces

In order to determine the internal forces within the structure, we will first calculate the external loads. At this stage of the design, we will only consider the self-weight. All other external loads can be neglected. Also, for simplification, the archlength of the vault can be set equal to its span. Draw a possible force flow and determine the magnitude of the inner forces as well as the support forces in the drawings below with the help of a force-diagram. Use the colours red for tension, blue for compression and green for the external forces.



Elevation (1:100)



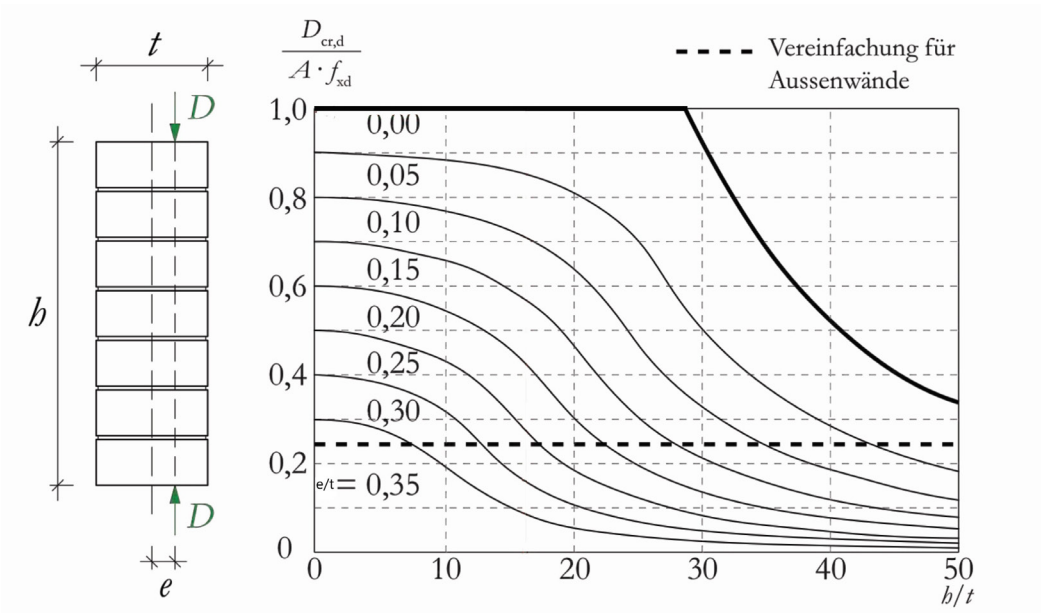
Floorplan (1:100)

Force diagramm (1cm:10kN)

1b Relations between form, force and material

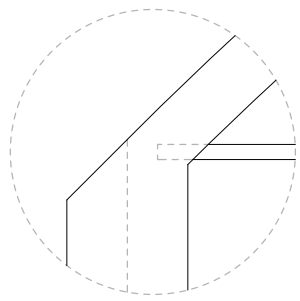
Hint:
«Skript TE 3-4» S.191 Knickdiagramm,
«Skript TE 3-4» S.190 Materialkennwerte,
Anhang «Formelsammlung»

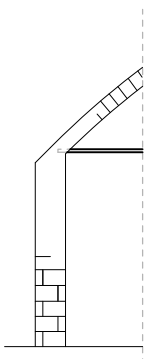
Evaluate if there is bukling risk in the wall by using the results of task 1a. The walls have hinge-support on both ends. The thickness (t) of the wall is 40 cm. And the force excentricity (e) is 10cm.



Buckling diagram for masonry

Detail 1





Can the walls resist a horizontal load applied in the center of the wall without any additional structural elements?